Shreyas Ghare

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Biomedical Engineering junior at Purdue University with research, leadership, and hands-on lab experience, seeking opportunities to advance skills in biomedical innovation and engineering design

Education

Purdue University (Main Campus) - Weldon School of Biomedical Engineering — West Lafayette, IN
Bachelor of Science in BME, plans to pursue a Master's — Class of 2028
GPA: 3.75 / 4.0

Experience

Research at HeaRT Laboratory — June - August 2026 Shumaker Research Building
Lab Research under Dr. Yash Chitalia at the University of Louisville
• Worked on a robot module to detect various surgical instruments through an ML model, and pick up the desired instrument and unlatch the device to open it, through a lead screw motor design and a custom-designed vacuum pickup tool

Research at Biomedical Devices Lab — May - August 2026 Shumaker Research Building
Lab Research under Dr. Guruprasad Giridharan at the University of Louisville
• Worked on a project designed to reverse the Fontan paradox, developing a novel, non-obstructive cavopulmonary assist device (CPAD) for patients with Fontan failure by testing novel impeller designs and running hemolysis comparisons against existing clinically approved blood pumps

University of Louisville Internship — 2022 - 2024 Clinical and Translational Research Building
Lab Research under Dr. Shirish Barve
• Conducted independent research on brain inflammation and alcohol's impact on the gut-brain connection through a BV-2 microglia cell line, gaining hands-on experience with biomedical techniques (Western Blot, PCR, assays) while completing biosafety and hazardous chemical training

Projects

Functioning ECG using an Arduino-MATLAB System — Spring 2026
• Circuits project using RC filters, operational amplifiers, and an Arduino microcontroller to take input readings through probes and process the data using MATLAB to display an ECG graph

Purdue EPICS - SPEEDWAY — Fall 2024 - Spring 2025
• Designed and assembled a functional wind tunnel for the Indianapolis Speed Museum using Fusion360, 3D printing for STEM education while strengthening teamwork, communication, and engineering design skills

Pathogenic Role of Aldose Reductase in Brain Inflammation — 2022-2024
• Investigated the effects of LPS (inflammation inducer) and EPAL (aldose reductase inhibitor) on microglia cells to study neuroinflammation, earning the Outstanding Achievement Award in In Vitro Biology from the Society for In Vitro Biology and membership in the SIVB community

Methods to Improve the Efficiency of Solar Panels — 2020-2023
• Designed and tested four solar panel models to improve efficiency, earning 2nd Place at the 2021 Kentucky Junior Academy of Science and 1st Place in Engineering at the 2021 Kentucky Academy of Science against undergraduate competitors; selected to present at the American Junior Academy of Science (AJAS) and the American Association for the Advancement of Science (AAAS)
• Published research abstract in the Kentucky Academy of Science scholarly journal: https://doi.org/10.3101/1098-7096-82.1.45

Additional

Additional Language Skills: Marathi - Professional Fluency Global Seal, Spanish - Working Fluency Seal of Biliteracy
Technical Skills: Programming Languages - Java, C, C++, Python, MATLAB, CAD Software - OnShape, Fusion 360, SolidWorks
Volunteer Experience: STEMY - non-profit bringing STEM education to underrepresented students
Indian Community Foundation - hosts events to celebrate Indian heritage
Interests/Hobbies: Guitar - 7 Years at Kentucky Arts Academy, Tennis - 5 Years during middle and high school
US Citizen